Stage 1 Report: Reproducing "Short Interest Trend Prediction with Large Language Models"

LLM Applications in Finance — Course Project, Stage 1 · 15 October 2026 ·

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Paper: Zhaomin Xiao, Yachen Cui, Zhelu Mai, Zhuoer Xu, Jiancheng Li. ICIAI 2024, Tokyo. doi:10.1145/3655497.3655500

1. Summary

Reproduction status: partial. We rebuilt the paper's dataset and protocol from public sources and measured two FLAN-T5 models on the same 500 test titles: FLAN-T5-large (780M parameters, CPU) and FLAN-T5-XL (3B parameters, Google Colab GPU). We did not run the paper's own models (FLAN-T5 11B, FLAN-UL2 20B, LLaMA-2 7B).

How to read this report: numbers labelled **paper** are copied from Xiao et al. (2024) and were not produced by us. Numbers labelled **ours** or **measured** come from our own prediction files (experiment_results.csv, predictions.csv).

- **Measured result (paper-faithful labels):** weighted F1 of 0.45–0.50 for FLAN-T5-large and 0.43–0.50 for FLAN-T5-XL across 0/1/5/10 examples. The paper's LLMs score 0.06–0.47 (recomputed from its Table 1).
- **Neither the paper's LLMs nor ours beat random guessing** (weighted F1 0.51). A simple TF-IDF + logistic regression classifier we trained reaches 0.60 on the same 500 titles (0.60 on the full test split).
- **The paper's labels measure short-sale volume, not short interest.** Its footnote 2 links to FINRA's daily short-sale volume files. We therefore also built a second label set from genuine FINRA short-interest reports; the two label sets agree on only 51.1% of titles.
- **True short-interest labels (set B):** weighted F1 0.48–0.51, within sampling noise of random guessing (0.50); every 95% interval includes it. TF-IDF + logistic regression reaches 0.57.
- **Validation:** our reconstructed labels match the gold labels of all four Figure 1 examples, and our class balance (55.7% increase) is within one point of the paper's (54.7%).

2. The paper in brief

The paper frames short interest trend prediction as binary classification: given one financial news title about company c, predict whether c's short interest goes up or down around the publication day.

Aspect	What the paper specifies
Input	One Benzinga news title; ASCII-only, at most 50 tokens
Label	increase if $S(t+1) - S(t-1) > 0$, decrease if < 0, ties dropped ($< 0.5\%$); $S(t)$ taken from FINRA daily short-sale files
Data	136,192 titles, 2009–2020, NASDAQ / NYSE / AMEX; 54.7% increase, 45.3% decrease
Split	Stratified 80/20 train/test
Models	FLAN-T5 11B, FLAN-UL2 20B, LLaMA-2-Chat 7B; Hugging Face 8-bit weights; one RTX 3090
Prompt (Figure 4)	"Read the news title below and determine how the short interest of {company} changes." + title + OPTIONS: 1. increase / 2. decrease + "ANSWER:"
In-context learning	0, 1, 5 and 10 examples before the question; inputs truncated at 2,048 tokens

Aspect	What the paper specifies
Metric	Per-class F1 and a “weighted average F1”; majority-class baseline
Headline claim	All LLMs beat the majority baseline; best is FLAN-UL2 10-shot with average F1 0.47

Not specified by the paper: how in-context examples are chosen, how model output is parsed, and which tokenizer counts title length.

3. Availability of original code and data

No official code or processed dataset could be found as of 14 October 2026, so the dataset was rebuilt from the public sources the paper names.

Where we looked	Result
The paper itself	No code or data availability statement; only source links (FINRA, Benzinga, a scraping tool)
ACM Digital Library page and Semantic Scholar	Closed access; no linked artifacts
GitHub and general web search	No repository for this paper
Hugging Face	First author's account (zhaominxiao) has one unrelated model and no datasets; co-authors have none
The news source the paper cites	Found: Kaggle dataset “Daily Financial News for 6000+ Stocks” (miguelaenlle, CC0), by the author of the Benzinga scraping tool in footnote 4

The Kaggle file `analyst_ratings_processed.csv` contains all four Figure 1 headlines verbatim, with matching tickers and dates. This is strong evidence it is the paper's news source, but it is not the authors' processed dataset.

4. Dataset reconstruction

Key finding. Footnote 2 links to FINRA's Daily Short Sale Volume Files, which FINRA describes as aggregated volume of off-exchange short-sale trades per day. Short interest is a different quantity: a snapshot of open short positions that brokers report twice a month. FINRA states the two data sets are not intended to equate. The paper's S(t) is therefore daily short-sale volume.

	Label set A: regsho_volume (paper-faithful)	Label set B: short_interest (true short interest)
Source	FINRA Reg SHO daily files, summed over Nasdaq, NYSE and Chicago TRFs and the ADF	FINRA consolidated short interest, exchange-listed stocks, twice monthly
S(t)	Shares sold short that day (off-exchange)	Open short position at a settlement date
Label rule	sign of S(next trading day) − S(previous trading day) around the news date	sign of SI(first report including news-day trades, T+2 settlement) − SI(previous report)
Period	Aug 2009 – Jun 2020	Dec 2017 – Jun 2020 (FINRA's public history starts 29 Dec 2017)
Labeled titles	1,313,978	372,512
Increase share	55.7%	51.9%
Ties dropped	0.02%	46 titles

Preprocessing: titles from `analyst_ratings_processed.csv`; ASCII-only and 1–50 tokens (FLAN-T5 tokenizer); news date = US Eastern calendar date of the timestamp; exact duplicate (title, ticker, date) rows removed. The company name in the prompt is FINRA's issue name where available (89% of titles), otherwise the ticker.

Validation. Label set A reproduces the gold labels of all four Figure 1 examples: Cobalt International Energy and Polaris = increase; Lindsay and Patterson Companies = decrease (4 of 4). Labels A and B agree on only 51.1% of the 370,845 titles that have both.

Statistic (paper Section 4)	Paper	Ours, label set A
Titles	136,192	1,313,978
Increase / decrease	54.7% / 45.3%	55.7% / 44.3%
Average title length, train / test	16.53 / 16.48 tokens	20.18 / 20.18 FLAN-T5 tokens (10.8 whitespace words)
Unique tokens, train / test	66,624 / 30,787	309,429 / 144,514 (whitespace)

The class balance matches within one point. Our dataset is about ten times larger from the same source; the paper does not explain how it arrived at 136,192 titles. About 46% of test titles also appear verbatim in the training split under other tickers (e.g., “Benzinga's Top Upgrades”), and 64% of repeated titles carry both labels, so the title alone cannot determine the label.

5. Experimental setup

Setting	Paper	Our measured runs
Model	FLAN-T5 11B, FLAN-UL2 20B, LLaMA-2-Chat 7B, 8-bit	google/flan-t5-large (780M), fp32, both label sets; google/flan-t5-xl (3B), not quantized, label set A only. Same FLAN instruction tuning as the paper's FLAN-T5
Hardware	One RTX 3090	FLAN-T5-large: 2 CPU cores. FLAN-T5-XL: Google Colab T4 GPU (PyTorch 2.11.0+cu130, Transformers 5.18.0).
Test set	Full 20% split	500 titles drawn from the 20% split (stratified, seed 42); identical for every shot setting
Prompt	Figure 4	Figure 4, word for word
Demonstrations	0/1/5/10; selection unstated	0/1/5/10 drawn at random from the training split, fixed per test title; each = full prompt + gold answer
Decoding	Completion after “ANSWER:”	Greedy, up to 5 new tokens; earliest of increase / decrease / 1. / 2. wins; anything else counts as wrong
Max input	2,048 tokens, truncated	2,048 tokens, truncated from the left so the question is never cut (longest prompt: 722 tokens)
Baselines	Majority class	Majority class; random guess at class prior; TF-IDF + logistic regression trained on the 80% split
Metric	Per-class F1, “weighted average F1”	Per-class F1, support-weighted F1, macro F1; 95% bootstrap intervals (2,000 resamples)
Seed	Not stated	42 (split, sample, demonstrations, random baseline)

6. Results

Important: Table 6.1 is copied from the paper and was not produced by us. Tables 6.2 and 6.3 contain only numbers we measured.

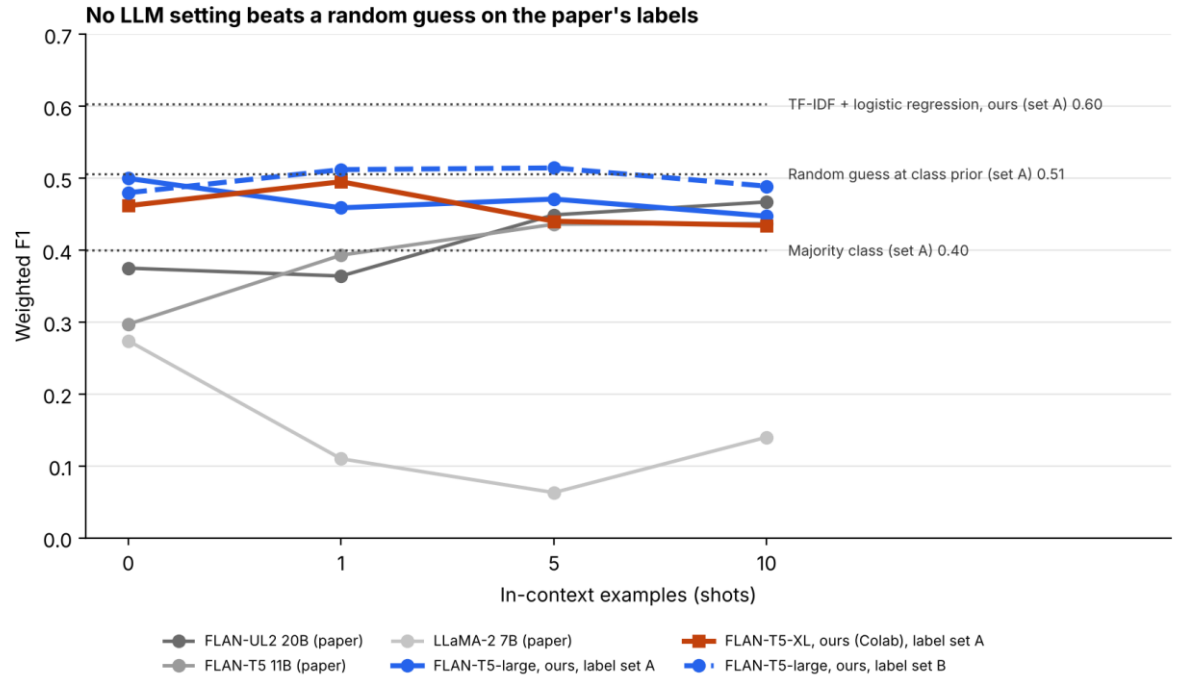


Figure 1. Weighted F1 by number of in-context examples. Paper values are recomputed from Table 1; ours are measured. Dotted lines are label-set-A baselines.

6.1 Paper results (reported, Table 1 of Xiao et al. 2024)

"Recomputed" = 0.547 × F1(increase) + 0.453 × F1(decrease), using the paper's own per-class numbers and class shares.

Model	Shots	F1 increase	F1 decrease	"Average" reported	Weighted F1 recomputed
Majority	–	0.71	0.00	0.35	0.39
FLAN-T5 11B	0	0.03	0.62	0.30	0.30
FLAN-T5 11B	1	0.23	0.59	0.40	0.39
FLAN-T5 11B	5	0.30	0.60	0.38	0.44
FLAN-T5 11B	10	0.31	0.59	0.44	0.44
LLaMA-2-Chat 7B	0	0.17	0.40	0.28	0.27
LLaMA-2-Chat 7B	1	0.06	0.17	0.08	0.11
LLaMA-2-Chat 7B	5	0.00	0.14	0.10	0.06
LLaMA-2-Chat 7B	10	0.00	0.31	0.39	0.14
FLAN-UL2 20B	0	0.18	0.61	0.37	0.37
FLAN-UL2 20B	1	0.16	0.61	0.36	0.36
FLAN-UL2 20B	5	0.34	0.58	0.45	0.45
FLAN-UL2 20B	10	0.39	0.56	0.47	0.47

6.2 Our measured results — label set A (paper-faithful short-sale-volume labels)

Model	Shots	n	F1 inc.	F1 dec.	Weighted F1 [95% CI]	Macro F1	% pred. inc.	Status
Majority class (always increase)	–	500	0.716	0.000	0.400	0.358	100%	complete
Random guess at class prior	–	262,796	0.556	0.442	0.506	0.499	–	complete
TF-IDF + logistic regression	–	500	0.614	0.590	0.603	0.602	–	complete
FLAN-T5-large (CPU)	0	500	0.524	0.469	0.500 [0.45, 0.54]	0.496	50%	complete
FLAN-T5-large (CPU)	1	500	0.534	0.364	0.459 [0.41, 0.50]	0.449	60%	complete
FLAN-T5-large (CPU)	5	500	0.560	0.359	0.471 [0.43, 0.52]	0.459	63%	complete
FLAN-T5-large (CPU)	10	500	0.590	0.267	0.447 [0.40, 0.50]	0.429	72%	complete
FLAN-T5-XL (Colab GPU)	0	500	0.398	0.542	0.462 [0.42, 0.51]	0.470	31%	complete
FLAN-T5-XL (Colab GPU)	1	500	0.626	0.330	0.495 [0.45, 0.54]	0.478	73%	complete
FLAN-T5-XL (Colab GPU)	5	500	0.662	0.160	0.440 [0.39, 0.49]	0.411	87%	complete
FLAN-T5-XL (Colab GPU)	10	500	0.674	0.132	0.434 [0.39, 0.49]	0.403	90%	complete

Every answer from both models was a valid option (0% unparseable). More examples push both models toward answering “increase”: FLAN-T5-large from 50% of titles at 0-shot to 72% at 10-shot, and FLAN-T5-XL from 31% to 90%. This raises F1 for increase and lowers it for decrease, so the weighted score does not improve. The larger model is not better: FLAN-T5-XL peaks at 0.495 (1-shot), still below the random-guess score of 0.506. The paper's FLAN models show the same shift in their per-class scores.

6.3 Our measured results — label set B (true short interest)

Model	Shots	n	F1 inc.	F1 dec.	Weighted F1 [95% CI]	Macro F1	% pred. inc.	Status
Majority class (always increase)	–	500	0.682	0.000	0.353	0.341	100%	complete
Random guess at class prior	–	74,503	0.515	0.479	0.497	0.497	–	complete
TF-IDF + logistic regression	–	500	0.580	0.551	0.566	0.566	–	complete
FLAN-T5-large (CPU)	0	500	0.476	0.484	0.480 [0.44, 0.52]	0.480	47%	complete
FLAN-T5-large (CPU)	1	500	0.531	0.492	0.512 [0.47, 0.56]	0.511	52%	complete
FLAN-T5-large (CPU)	5	500	0.575	0.450	0.514 [0.47, 0.56]	0.512	61%	complete
FLAN-T5-large (CPU)	10	500	0.620	0.348	0.489 [0.44, 0.54]	0.484	75%	complete

All four shot settings completed, with 0% unparseable answers. As on set A, more examples shift answers toward “increase” (47% at 0-shot, 75% at 10-shot), and no setting is distinguishable from a random guess.

7. Why our numbers differ, and problems in the paper

7.1 Differences between our run and the paper

- **Model size:** FLAN-T5-large (780M) and FLAN-T5-XL (3B) vs FLAN-T5-XXL (11B) and FLAN-UL2 (20B). The paper's FLAN-T5 11B answers “decrease” almost every time at 0-shot (F1 0.03 for increase); our XL leans the same way at 0-shot (69% “decrease”), while our large model splits its answers about evenly. Going from 780M to 3B did not raise the weighted F1.
- **Sample:** 500 test titles, so each weighted F1 carries a 95% interval of roughly ±0.04–0.05; most differences between our shot settings are within that noise.
- **Dataset size:** 1.31M labeled titles vs 136,192; the paper's subset is unknown, so the test titles differ.
- **Unstated choices:** demonstration selection, output parsing and the length tokenizer had to be chosen by us.
- **Metric:** the paper's “Average” column is not computed consistently (7.2, item 2), so only its per-class F1 values can be compared directly.

7.2 Problems found in the paper itself (checkable from the PDF)

1. **The label is short-sale volume, not short interest.** Footnote 2 points to FINRA's daily short-sale volume files, which count shares sold short each day off-exchange.
2. **The “Average” column is not consistently a weighted F1.** Recomputing it from the per-class numbers matches only 8 of 13 cells within 0.01. Mismatches: Majority (0.39 vs 0.35 reported, which equals the macro average), FLAN-T5 5-shot (0.44 vs 0.38), LLaMA-2 1-shot (0.11 vs 0.08), 5-shot (0.06 vs 0.10) and 10-shot (0.14 vs 0.39).
3. **“All LLMs outperform the majority baseline” contradicts Table 1:** LLaMA-2 scores 0.08 (1-shot) and 0.10 (5-shot) against a majority score of 0.35.
4. **No reported LLM result beats random guessing.** With 54.7% / 45.3% classes, guessing at the base rate gives weighted F1 = $0.547^2 + 0.453^2 \approx 0.50$; the best LLM cell is 0.47.
5. **Some results reflect output-format failures.** FLAN-T5 0-shot (0.03 / 0.62) equals always answering “decrease” (F1 = 2 × 0.453 / 1.453 = 0.62). LLaMA-2 at 0.00 / 0.14 fits mostly unparseable answers.
6. **Text and Figure 1 disagree.** The introduction says downgrades make short sellers cover and short interest then likely increases; covering lowers short interest, and Figure 1 labels that example “decrease”.

8. Limitations and Stage 2 plan

Limitations

- Smaller models than the paper (780M and 3B vs 7B–20B). FLAN-T5-XL was run on label set A only.
- 500-title evaluation sample rather than the full test split.
- Dataset size differs from the paper for reasons the paper does not state.
- Demonstration selection, output parsing and length tokenizer are our choices.
- Many titles are market-wide lists attached to dozens of tickers, so the same title appears in train and test with different labels.
- Label set B covers only Dec 2017 – Jun 2020 because FINRA's free short-interest history starts there.

The short-sale-volume vs short-interest limitation

This is the most important limitation of the paper, and therefore of any faithful reproduction of it.

- **Short-sale volume** (what the paper's label measures): the number of shares sold short on one day, counted from off-exchange trade reports. It changes every day and rises and falls with ordinary trading activity.
- **Short interest** (what the paper says it predicts): the total number of shares currently held short. It is a stock of open positions, reported only twice a month.

- A rise in daily short-sale volume does not mean short interest rose: many short sales are closed the same day, and covering trades are not counted as short sales.
- In our data the two labels agree on only 51.1% of titles, close to a coin flip. Results on label set A therefore say little about short interest; label set B is the valid test of the paper's stated task.

Stage 2 plan

1. Run LLaMA-2-7B-Chat in 8-bit (free Colab T4) and FLAN-T5-XL on label set B, on the same 500 titles; FLAN-T5-XXL and FLAN-UL2 on an A100 if available. (FLAN-T5-XL on label set A is done.)
2. Re-score with option ranking (SCORING = "rank") to separate model knowledge from output-format failures.
3. Fine-tune with LoRA, as the paper proposes in its future work, and compare against the TF-IDF baseline.
4. Treat label set B (true short interest) as the main task, with set A kept for comparison with the paper.

9. Presentation outline

1. **Title** — paper, course, team.
2. **Problem** — what short interest is; task: one news title → increase / decrease.
3. **Paper's method** — Benzinga + FINRA data, Figure 4 prompt, three LLMs, 0/1/5/10-shot, F1.
4. **Paper's claims** — Table 1; best FLAN-UL2 0.47; “examples help”.
5. **Reproduction challenge** — no code or data released; how we found the Kaggle source and FINRA files.
6. **Key finding** — labels are short-sale volume, not short interest; two label sets; 51% agreement.
7. **Validation** — Figure 1 labels 4 of 4; class balance 55.7% vs 54.7%.
8. **Our setup** — FLAN-T5-large (CPU) and FLAN-T5-XL (Colab), 500 titles, seed 42, greedy decoding, baselines.
9. **Results** — Figure 1 of this report: no LLM beats random; TF-IDF reaches 0.60.
10. **Analysis** — examples shift label bias; inconsistencies in the paper's Table 1.
11. **Limitations and Stage 2 plan.**
12. **Demo** — the Colab notebook, then questions.

10. Sources and deliverables

Sources

- Xiao et al. (2024). Short Interest Trend Prediction with Large Language Models. ICIAI 2024. https://doi.org/10.1145/3655497.3655500
- Kaggle: Daily Financial News for 6000+ Stocks (miguelaenlle). https://www.kaggle.com/datasets/miguelaenlle/massive-stock-news-analysis-db-for-nlpbacktests
- FINRA: Daily Short Sale Volume Files. https://www.finra.org/finra-data/browse-catalog/short-sale-volume-data/daily-short-sale-volume-files
- FINRA: Short Interest – What It Is, What It Is Not. https://www.finra.org/investors/insights/short-interest
- FINRA consolidated short interest files. https://cdn.finra.org/equity/otcmarket/biweekly/

Deliverable files

File	Contents
stage1_reproduction.ipynb	Colab notebook: data download/rebuild, preprocessing, baselines, prompting, inference, F1 evaluation
experiment_results.csv	All measured results with status (complete / incomplete); no paper numbers
predictions.csv	All 6,000 measured predictions (FLAN-T5-large and FLAN-T5-XL): model, label set, shots, where run, title, true label, predicted label, raw output
experiment_config.json	Model, prompt, decoding, seed, dataset sources, split and test-set sizes
sitp_dataset.parquet	Rebuilt labeled dataset with both label sets (1,334,522 rows)